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Professor Wolfgang Polifke
Technical University of Munich (TUM)
Professorship of Thermofluid Dynamics
Munich, Germany

May 5, 2026

Subject: Application for Doctoral Research - Physics-Informed Digital Twins

Dear Professor Polifke,

I am writing to formally submit my application for a doctoral position under your supervision at the Technical University of Munich. As an AI Engineer specializing in hybrid physics-data modeling, I have closely followed your pioneering research on Physics-Informed Deep Learning for thermofluid dynamics, and I am eager to contribute to this field.

Currently, I am concluding my state engineering thesis at **Sonatrach**, where I have been tasked with building a **Generative Digital Twin** for a fleet of LM2500 gas turbines. My research address the "reality gap" by embedding thermodynamic constraints directly into neural network architectures using **PINN** loss functions. I have found that this hybrid approach is essential for maintaining physical consistency in predictive maintenance scenarios where data can be sparse or noisy.

I am particularly interested in your work on modeling turbulent flows and combustion instabilities. My experience with **Graph Neural Networks (GNNs)** for modeling structural dependencies, combined with my proficiency in sequential modeling using **Mamba-SSM**, allows me to approach thermofluid problems from a high-performance system perspective. I am keen to investigate how these state-of-the-art architectures can be integrated into your thermofluid models to improve extrapolation capabilities in extreme operating conditions.

TUM is my first choice for a PhD due to its world-class research in Engineering and Informatics. I am a highly motivated and autonomous researcher, and I believe my industrial background at Sonatrach has prepared me to tackle the rigorous challenges of your laboratory.

I have attached my academic transcripts, my CV (noting that ESI-SBA is an H+ recognized institution), and a research proposal for your review. I would be honored to discuss my potential contributions to your group during an interview.

Thank you for your time and consideration.

Sincerely,

Ismail AISSAOUI